Combatting Space Debris: Modeling Laser Ablation to Minimize Plasma Shielding through Changing the Pulse Shape of a High-Power High-Frequency Laser

Background

Space debris: Space debris is made up of defunct technology and pieces of spacecraft traveling at several kilometers per second (Haridim et al., 2026). The accumulation of space debris in low Earth orbit poses a threat to current and future missions to space. One of the mitigation strategies for space debris is the use of space-based lasers to vaporize the debris.

Plasma shielding

Laser ablates a material → plasma plume is created

In a high frequency laser, the next pulse may hit the material before the plasma plume can fully dissipate, which leads to the incoming laser pulse to be deflected or absorbed by the laser. This phenomenon is called plasma shielding and can decrease the effectiveness of laser ablation.

Pulse shape: The pulse shape of a laser shows the distribution of energy over time for each pulse. Changing the pulse shape can lead to more effective distributions of enegy for limiting the puildup of plasma, thus limiting plasma shielding.

Question

Which pulse shape most effectively carries out laser ablation as seen in lower plasma shielding and higher ablation depth?

Methods

Independent variable: Pulse shape (rectangular pulse, gaussian pulse, exponential pulse, sawtooth pulse).

Dependent variables: Ablation depth and energy deflected.

Constants: Total power used, laser characteristics, laser frequency, space debris characteristics, number of pulses per trial.

Control: N/A

MATLAB application was used to model laser ablation of aluminum using four different pulse shapes. A MATLAB was written with equations representing pulse shape, heat flux, material ablation, and energy transfer and more. The variables in these equations were then replaced with values calculated from the defined laser and aluminum characteristics. The code modeled the heat movement through a cube of aluminum. The aluminum was split into cross sections to most effectively simulate heat movement through the cube. The ablation depth and total energy deflected, which shows plasma shielding, was calculated by the model and then recorded for analysis.

Data / Results

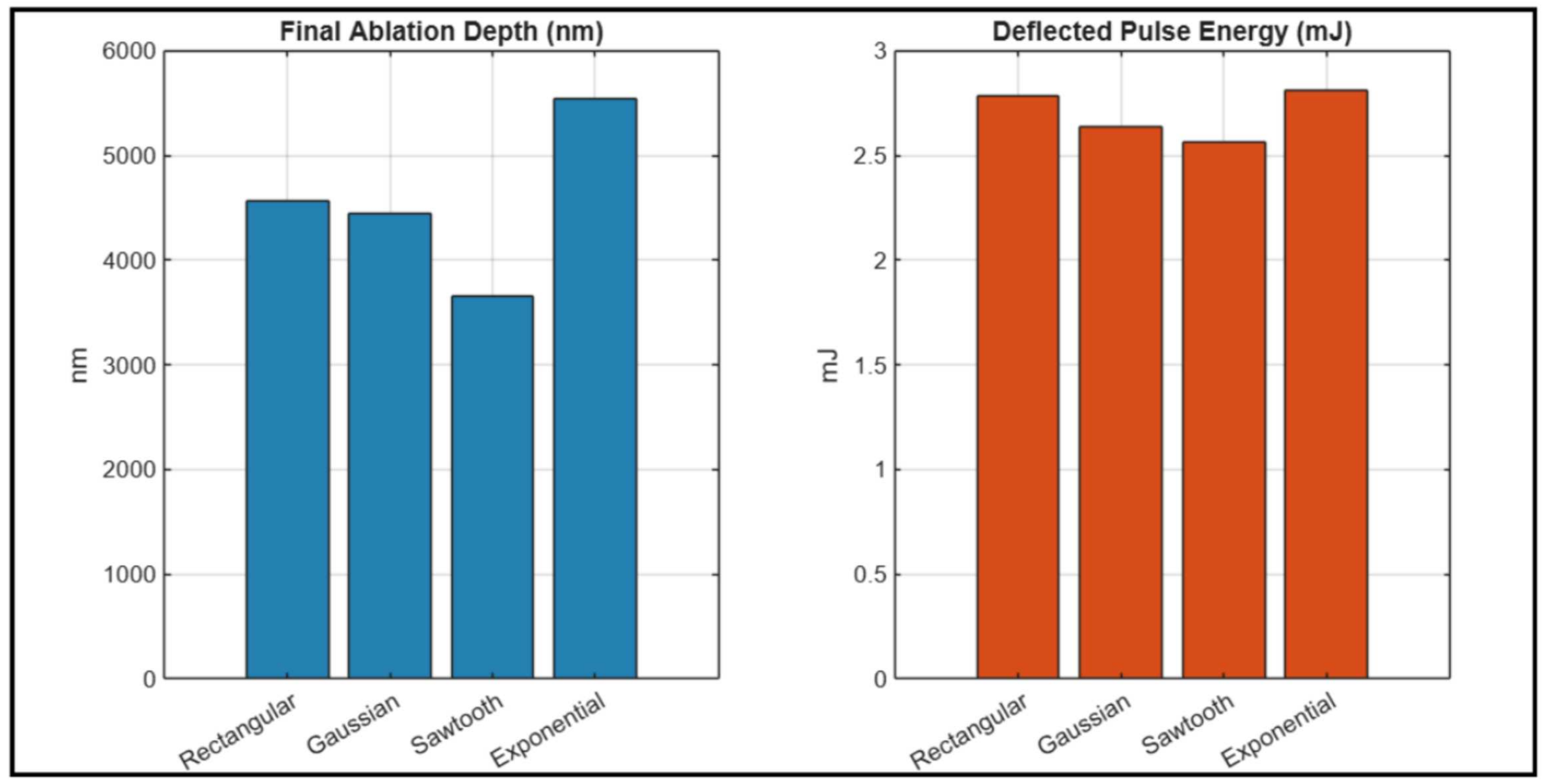


Figure 1: Ablation depth and deflected energy data for each pulse shape

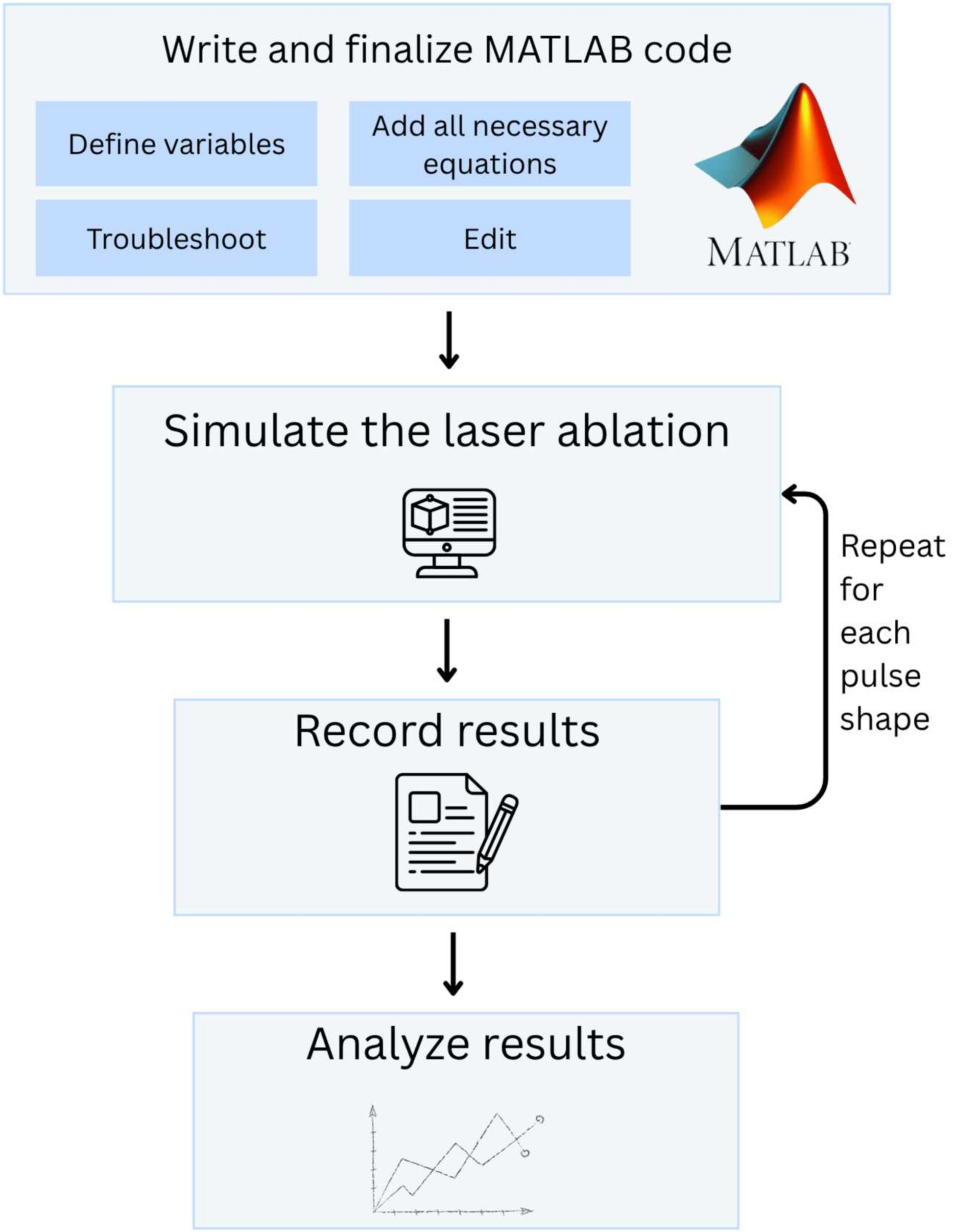
These graphs depict the data from the MATLAB code used to calculate the ablation depth and plasma shielding (deflected energy) of four different laser pulse shapes.

The sawtooth pulse had the lowest plasma shielding but also had the lowest ablation depth as well.

The Total Ablation Depth, Deflected Energy, and Ratio Between the Ablation Depth and Deflected Energy for Each Pulse Shape			
Pulse shape	Total ablation depth (nm)	Deflected energy (mJ)	Ratio (ablation depth: deflected energy)
Rectangular	4567.89	2.79	1637.24
Gaussian	4446.6	2.64	1684.32
Sawtooth	3658.26	2.56	1429.01
Exponential	5539.77	2.81	1971.45

Figure 2: ablation depth, deflected energy data, and ablation depth:deflected energy ratio for each pulse shape

Method Visual



Results & Conclusions

Data: The data obtained from this study included the total ablation depth and deflected energy from the modeled laser ablation of each of the four pulse shapes.

Results:

- Figure 1 shows the sawtooth pulse showed the least deflected energy/plasma shielding.
- Figure 1 shows the exponential pulse showed the highest total ablation depth.
- Figure 2 shows the exponential pulse was determined to lead to the most effective plasma shielding.
  - Had the largest ratio between total ablation depth and deflected energy.

Data analysis:

- Statistical tests could not be carried out
  - Because the data from this study came from code, only one data point was produced per pulse shape.
- Because only one data point was created per pulse shape, standard deviation error bars could not be created as well.
- The reliability of the code was determined through perfroming a verification experiment.

Results & Conclusions

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Pulse Shape     | Ablation (nm)
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Rectangular     | 23.50
Gaussian        | 36.25
Sawtooth        | 32.36
Exponential     | 31.32
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Figure 3: Ablation depths from the verification experiment

Verification experiment:

- Determine the validity of the code.
- Used a study of physical aluminum laser ablation.
  - Switched the laser parameters in the code and recorded the ablation depth of the aluminum.
  - Needed an ablation depth between 30nm and 60nm with the Gaussian pulse for the validity of the code to be supported.
  - Figure 3 shows that the ablatied depth was 36.25 nm, thus the code was seen as valid.

Limitations & Future Research

Limitations

- One limitation of this study is that it only studied aluminum plasma shielding.
  - Space debris is made up of a wide range of materials.
- The aluminum in the MATLAB script was a perfect cube.
  - In reality, space debris is rarely ever a perfect cube.
- The MATLAB code shows 2D ablation rather than modeling 3D heat dynamics.

The limitations in the model were not thought to have had any detrimental effects to the data. This is because to study pulse shape, 2D heat modeling has been found to be adequate in prior literature. The simplified model allowed the script to be made with little prior experience with coding.

Future Research

- Carrying out the model with different target metals and shapes
- Investigating a wider range of pulse shapes

References

Haridim, M., Shaked, A., Cohen, N., & Gavan, J. (2026). Challenges of Space Debris Detection, Tracking, and Monitoring in Near-Earth Orbit: Overview of Current Status and Mitigation Strategies. Information, 17(3), 253. https://doi.org/10.3390/info17030253

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